

Assignment : Windows Ransomware Early Detection

Group ID : 47

1. Objective

The primary goal of this project was to implement the feature extraction methodology proposed in the RENTAKA framework(as per our chosen research paper). The objective is to detect Windows cryptographic ransomware early in its lifecycle—specifically during the "setup" phase—before the irreversible file encryption process begins.

2. Dataset and Pre-processing

We evaluated our pipeline using the KHAS-CCIP VirusShare dataset. Processing this real-world dataset presented a significant engineering challenge due to its "long sequence" format. The raw dataset contained 14,616 samples. Expanding these API sequences into a dense matrix initially caused memory allocation failures.

To solve this, we engineered a memory-safe dataloader using a C-optimized [CountVectorizer](#) to generate a sparse matrix. This allowed us to successfully extract the top 2,000 unique API calls without crashing the system.

Following the RENTAKA algorithm's logic, we scanned for encryption-related API imports (the "boundary markers"). We identified and removed 11 cryptographic APIs, leaving a final feature matrix of 14,616 samples and 1,989 pre-encryption API features.

3. Experimental Results

The dataset reflects a realistic, highly imbalanced zero-day environment: 115 ransomware samples hidden among 14,501 other malware types. We trained five classifiers using 10-Fold Stratified Cross-Validation. The entire training and evaluation pipeline completed in approximately 560 seconds.

Classifier	Our Accuracy (%)	Paper Baseline (%)	Difference (Δ)	TPR	FPR	F1-Score
kNN	99.7947	96.07	+3.73	0.783	0.000	0.857
Random Forest	99.6648	96.39	+3.27	0.835	0.002	0.797

SVM	99.5758	97.05	+2.53	0.835	0.003	0.756
J48 (Decision Tree)	99.5484	94.75	+4.79	0.835	0.003	0.744
Naive Bayes	79.2967	80.98	-1.69	0.870	0.208	0.062

4. Discussion and Conclusion

Our implementation successfully outperformed the original paper's accuracy baselines in four out of five models. Interestingly, while the original paper found SVM to be the best classifier, our real-world dataset run showed that **k-Nearest Neighbors (kNN)** achieved the highest overall accuracy at 99.79%.

It is important to note the impact of the dataset's class imbalance. Because there were only 115 true ransomware samples compared to 14,501 non-ransomware samples, the models heavily favored the majority class, resulting in slightly lower True Positive Rates (TPR) and F1-scores compared to the perfectly balanced dataset used in the original study. However, the near-zero False Positive Rates (FPR) across kNN, Random Forest, and SVM prove that our memory-safe implementation of the RENTAKA feature reduction strategy is highly viable for enterprise endpoint protection.

GitHub Repo : https://github.com/Kumar-Anurag-4505/CyberSec_assign3

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